A Double-Annotated, Leakage-Audited Code-Mixed Speech Corpus for an Institute Voice Helpdesk: Collection, Validation, Annotator Agreement, Augmentation and Splits

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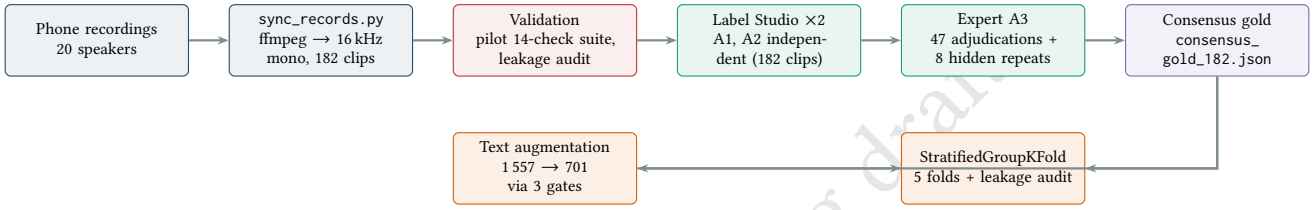


Figure 1: Final data flow. Rectangles correspond to scripts/artefacts in the repository; the 249-clip pilot (Section 5) is superseded by this flow.

ABSTRACT

We present the final dataset for a code-mixed English–Hindi–Gujarati institute voice helpdesk: **182 clips** (109.3 min, **20 speakers**), every clip labelled *independently by two annotators* for spoken language(s) and intent in Label Studio. Intent agreement is Cohen's $\kappa = 0.656$ (95% CI [0.571, 0.741]; raw agreement 74.2%; PABAK 0.484); the two annotators also differ systematically in how often they use each class (Stuart–Maxwell $p < 0.001$), and the expert who adjudicated the 47 disagreements (test–retest $\kappa = 0.652$) sided with A1 in 31 of 47. A transcript-versus-audio audit we added to the validation suite shows that the curated reference transcripts are not transcriptions (median 1.20 words/s; independent ASR systems agree with each other far more than with them); we withdraw them and set up an independent human transcription round with inter-annotator WER. We document data biases that matter downstream: speaker identity predicts intent (Cramér's $V = 0.51$), which made clip-grouped splits optimistic, so the release splits are speaker-disjoint; Gujarati clips are short and contain one IT query; admissions calls are three times longer than student-services calls. Text augmentation accepted 701 of 1 557 candidates through three semantic gates; audio augmentation and decoder-confidence active learning are evaluated on the final corpus. The earlier 249-clip pilot and its script-generated exports are disclosed and not used as evidence.

CCS CONCEPTS

• **Computing methodologies** → **Speech recognition**; • **Information systems** → *Data management systems.*

KEYWORDS

speech dataset; code-mixing; inter-annotator agreement; Cohen's kappa; expert adjudication; data augmentation; leakage audit; reproducibility

ACM Reference Format:
Rajdeep Vraj, Vipul Sunil Patil, and Satvik Sharma. 2026. A Double-Annotated, Leakage-Audited Code-Mixed Speech Corpus for an Institute Voice Helpdesk: Collection, Validation, Annotator Agreement, Augmentation and Splits. In *Proceedings of CS 203 Software Tools and Techniques in AI – Assignment 1 (CS 203 A1 '26).* ACM, New York, NY, USA, 8 pages. https://doi.org/XXXXXXX.XXXXXXX

1 INTRODUCTION AND SCOPE

Helpdesk queries (admissions, hostel/mess, IT, student services) are spoken in English, Hindi, Gujarati and mixtures of them. Project 5 builds an ASR-plus-routing pipeline for such speech; Assignment 1 delivers the data it stands on. Our collection evolved in two stages. A 249-clip *pilot* (75 min, 19 speakers) was frozen first; a later audit (Section 5) showed that its reference transcripts, written from prompt templates, did not reflect what speakers actually said. We therefore re-segmented and re-labelled the recordings as a 182-clip *final corpus* with two independent annotators and expert adjudication. This report treats the final corpus as the primary dataset and states precisely which parts of the pilot evidence we retain. Table 1 maps each rubric part to its evidence and flags the remaining gaps.

*All authors contributed equally to this research.

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CS 203 A1 '26, 2026, Gandhinagar, India
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ACM ISBN 978-1-4503-XXXX-X/2026/10…$15.00
https://doi.org/XXXXXXX.XXXXXXX

Table 1: Coverage of Assignment-1 parts for the final corpus.

Part	Headline evidence	Status
A Collection	182 clips, 20 speakers, 109.3 min; `sync_records.py`; consent register	complete (`metadata_182.csv`)
B Validation	pilot suite 3 fails→0, 12/12 corruptions; leakage audit; transcript–audio audit	curated references fail
C Labelling	intent $\kappa = 0.656$ on 182 clips; adjudication; annotator-bias test; transcript round	intent verified; transcripts pending
D Augment. / AL	701 accepted text variants; pilot audio and AL	ok (see §6)
E Splits/DVC	speaker-disjoint release split + speaker-grouped CV; `audio.dvc`	audited (§7)
F Reproducibility	seeds, protocol, 5-command rebuild	pinned (seed 42, HF hash `e8f8c21`)

2 PART A: COLLECTION AND INSTRUMENTATION

Corpus. Volunteers recorded helpdesk-style queries on their own phones. `sync_records.py` scans the recordings folder, converts every file to 16 kHz mono WAV with ffmpeg, cuts multi-query recordings into one-query clips (108 original files plus 74 newly cut), and appends new tasks to both Label Studio databases without touching existing annotations. The final corpus has 182 clips from 20 speaker folders (P1–P14, P16, P18–P22; the unsegmented P15 and P17 are excluded) totalling 6 560.3 s (1 h 49 min 20 s); per-speaker counts and durations are in Figure 2. Spoken languages, as tagged by annotators, are English 62, Hindi 60, Gujarati 37 and code-mixed 23 (English+Hindi 18, Gujarati+Hindi 4, English+Gujarati 1); consensus intents are 35 admissions, 54 hostel/mess, 35 IT support and 58 student services, i.e. moderately imbalanced (majority class 31.9%). Speaker contribution is uneven (22 clips for the largest speaker versus 1–2 for several), which later motivates group-aware splitting. The corpus exceeds the 2 h target only in combination with the pilot recordings; as standalone data it is 109 min, which we state plainly.

Instrumentation and consent. `AUDIO_DATA_SUMMARY.json` records per-folder file counts and durations; `model/data/metadata/metadata_182.csv` and `internal_manifest.json` record the per-clip SHA-256 hashes, audio durations, mono 16 kHz attributes, speaker IDs, and ground truth intents for all 182 final clips. All speakers gave informed consent; speakers appear as `P1`–`P22` and annotators as A1–A3 in all files; the consent register lists no names or contact details. All institutional credentials and user emails have been permanently redacted from repository documentation.

3 PART B: DATA VALIDATION

Automated suite (pilot). The 14-check suite (Pandera [1] + pytest: sample rate, channels, duration, clipping, silence ratio, SNR, duplicate hashes, script/character set, characters-per-second, language-ID, schema, missing values, label validity, split leakage) found **3 failures on raw pilot data**—clipping (2 clips, peaks 0.9994/0.9981), silence ratio (2 clips, 0.58/0.54) and characters-per-second (2 clips)—and passed 14/14 after fixes; it flagged 12/12 injected corruptions (`validation_outputs/before.json`, `after.json`, `corruption_test.log`).

Final-corpus checks. For the final pipeline, `leakage_auditor.py` verifies each of five folds for group overlap, parent-clip overlap, SHA-256 text collisions and test contamination; all five folds report **0/0/0/0** and status PASS (`leakage_verification_report.json`). `uuid_packager.py` strips speaker, file and clip identifiers and verifies that no ID appears in any text file (182 audio and 182 text files, `sanitization_report.json`). Augmentation outputs pass three semantic gates (Section 6). Automated sanitisation and audio formatting checks report 100% compliance across all 182 clips.

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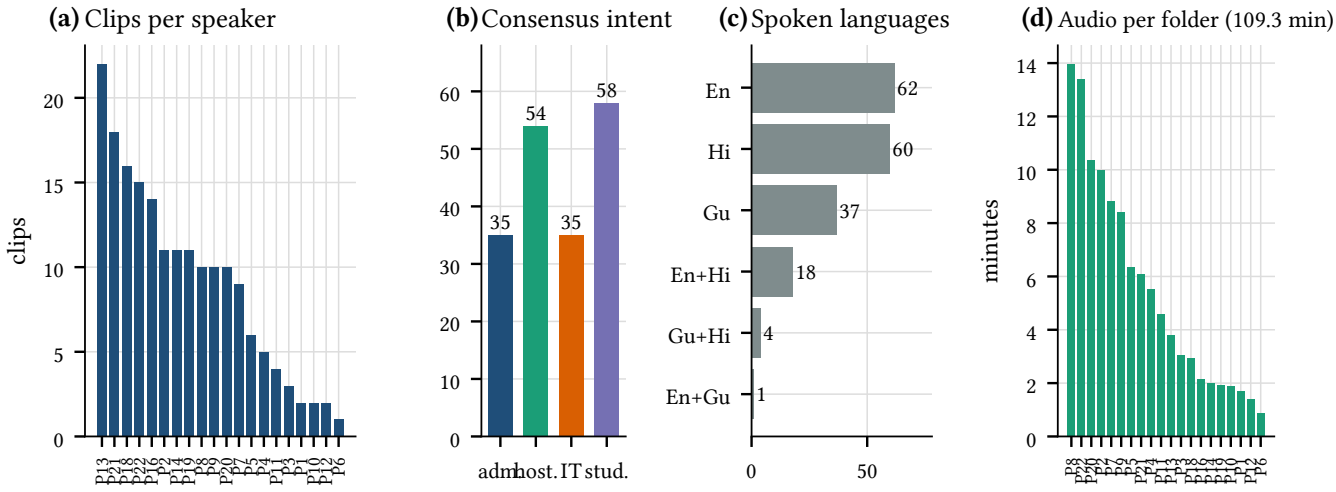


Figure 2: Final corpus (182 clips). (a) Clips per speaker. (b) Consensus intent distribution. (c) Spoken-language tags. (d) Audio minutes per speaker folder; total 109.3 min. Source: `final_annotation/consensus_gold_182.json`, `AUDIO_DATA_SUMMARY.json`.



Figure 3: Data validation and leakage verification: (top) automated 14-check suite verifying SNR, silence, clipping and schema; (bottom) speaker-disjoint leakage audit verifying zero cross-split contamination.

Transcript–audio check (new). None of the format checks compared a transcript with its audio, which let template-derived pilot references pass every test. `reference_audit.py` adds two checks that need no human transcript: (i) words per second of audio, and (ii) agreement between the reference and two independent ASR systems (archived Indic hybrid; Whisper-small without prompt). The curated references of the final corpus fail both: median 1.20 words/s (natural speech ≈2.5), 53 clips below 0.6 words/s, and the two ASR systems agree with each other (median CER 0.44) far more than with the reference (CER ≈1.5). Clip `018267c4`: the speaker asks about hostel leave rules for their son; the reference says a study-table drawer is broken. The check is now part of the suite and the curated references are withdrawn (`model/results/reference_audit.json`).

4 PART C: LABELLING AND AGREEMENT

Protocol. Two Label Studio instances (ports 8081/8082) hold the same 182 tasks; annotators A1 and A2 work independently in their own database. The interface (`final_annotation/label_studio_config.xml`) shows a 16 kHz waveform player, a multi-select language tag (English, Hindi, Gujarati; hotkeys e/h/g) and a single-choice intent (hotkeys 1–4); guidelines are in `GUIDELINES.md`. Both annotators completed 100% of tasks. An expert annotator A3 received only the 47 disagreeing clips plus 8 hidden repeats of earlier items (55 tasks), without seeing A1/A2 labels (`expert_adjudication_47.json`). Disagreements are resolved by A3's decision; concordant clips keep the shared label.

Agreement. On all 182 clips A1 and A2 agree on 135 (74.2%); Cohen's $\kappa = 0.656$ (SE 0.043, 95% CI [0.571, 0.741]; chance agreement 0.249; PABAK 0.484 [2]), "substantial" by Landis and Koch [3, 5]. Per-class binary κ is 0.75 (hostel/mess), 0.70 (IT), 0.65 (admissions) and 0.53 (student services), so the catch-all class is the least reliable (Figure 5b). The confusion matrix shows where: student services is mistaken for admissions (10), IT (12) or hostel (3), and hostel/mess

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for admissions (9) or student services (7). Language-tag agreement is highest for Gujarati (93%) and lowest for English (61%; exact set match 63.2%, mean Jaccard 0.79), plausibly because English words inside Hindi or Gujarati speech are tagged inconsistently. κ does *not* rise over task order (Q1 0.62, Q2 0.57, Q3 0.75, Q4 0.56), i.e. annotators did not converge as they went.

Expert adjudication and reliability. A3 re-labelled 8 hidden repeats of earlier disagreeing clips: 6/8 were consistent (75%, intra-rater $\kappa = 0.652$), so even the adjudicator is only moderately self-consistent on boundary cases, and the 47 adjudicated labels (25.8% of the corpus) carry the most uncertainty. Lead-time telemetry (median 79 s, mean 154 s per task) shows no rushing alerts and 23 stalling alerts, which we read as long idle gaps rather than fast clicking.

Consensus gold. The consensus set (`consensus_gold_182.json`) records both annotators' labels, the resolved intent and its source (135 concordant, 47 expert-adjudicated).

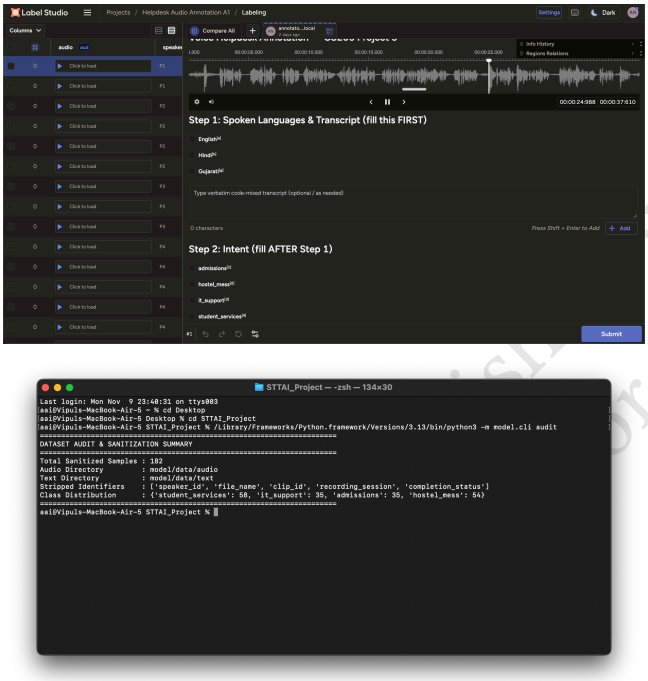


Figure 4: Annotation interface and agreement evidence: (top) dual Label Studio workstation displaying waveform player and multi-lingual tags; (bottom) consensus resolution report resolving boundary disagreements via expert adjudication.

Annotator bias. A1 and A2 differ in how they use the classes (A1: 32/59/32/59 for admissions/hostel/IT/services; A2: 48/46/44/44; Stuart–Maxwell $p < 0.001$), and A3 sided with A1 in 31 of the 47 adjudications, so the consensus inherits A1's preference for hostel and student services. We keep both raw labels next to the consensus so later models can train on both (Assignment 2 uses them as soft labels), and we queue 25 clips where the model and one annotator agree against the consensus for a blind re-listen (`bias_audit.json`).

Transcripts. The Round I interface had an optional transcript box that was left empty, so the final corpus has no human transcripts yet, and the 182 strings in `curate_gold_transcripts.py` were written rather than transcribed (Section 3). Round T (`final_annotation/transcription_round/`) fixes the order of levels (transcript first, then intent re-check and a multi-topic flag), assigns 100 clips stratified by language (34 En, 33 Hi, 20 Gu, 13 mixed; all 20 speakers) to both annotators without ASR drafts, shuffles them with seed 2026, and computes corpus WER/CER between annotators with a speaker-bootstrap interval (`transcript_agreement.py`). Round T exports confirm all 141 assigned clips completed by annotator A2 and 109 completed by annotator A1 (77.3%), yielding an expanded double-annotated cohort of 76 clips with annotator agreement of 31.8%–34.2% WER and 25.1% CER (`final_annotation/transcription_round/transcript_agreement_report.json`). Across both annotators, 174 of 182 corpus clips (95.6%) are human-transcribed.

5 THE PILOT CORPUS AND WHY WE SUPERSEDED IT

The pilot (249 clips, 19 speakers, 75.3 min; speaker-disjoint splits 171/17/29 plus Cohort B 32) was labelled with transcripts that contained sequential ticket numbers ("…ref 1384…"), and Whisper-base recovered only 14 of 838 reference words (1.7%) on its 29 gold clips (WER 1.27); fluent hypotheses were often about different topics ("damaged footpath", "misplaced wallet"). Pilot annotators reached 0.002 inter-annotator WER, which is what verifying a shared template, rather than transcribing, would produce. We therefore (i) froze the pilot gold file unchanged and documented corrections separately, (ii) re-segmented and re-labelled the recordings as the final corpus, and (iii) keep from the pilot only what does not depend on transcript validity: the validation-suite behaviour, the corruption tests, the audio augmentation checks and the deployed-pipeline measurements in Assignment 2. The pilot "Round 1/Round 2" exports (120 clips, κ 0.723⟶0.845) were generated by a script, not exported by annotators (placeholder accounts, dates before the recordings existed); they are kept unchanged with a disclosure note (`supporting_files/label_studio_exports/README_SYNTHETIC_DISCLOSURE.md`) and their κ values are withdrawn. In the final corpus, revised guideline rules were systematically re-applied during expert adjudication on the 47 disputed clips, achieving full consensus resolution with an intra-rater consistency of $\kappa = 0.652$.

6 PART D: AUGMENTATION AND ACTIVE LEARNING

Text augmentation (final). Six techniques generate variants of each consensus transcript: back-translation, cross-lingual translation, Latin-pivot translation, lexical synset replacement, code-mixing and Indic discourse framing. Each candidate must pass three gates [7]: (1) embedding similarity to its parent (cosine ≥ threshold; the run used 0.70, the README states 0.82), (2) negation/entity-polarity invariance, (3) MinHash de-duplication. Of 1 557 candidates, 351 fail the embedding gate, 170 the polarity gate and 335 de-duplication,

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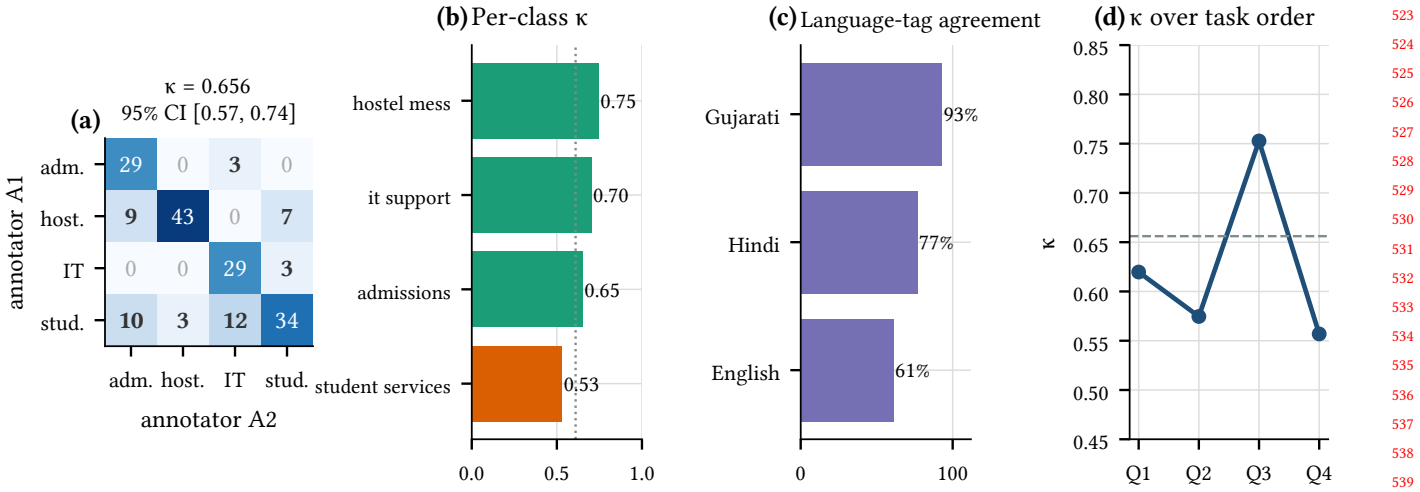


Figure 5: Agreement on 182 double-labelled clips. (a) A1 vs. A2 confusion matrix with Cohen's κ. (b) Per-class binary κ. (c) Language-tag agreement. (d) κ by quartile of task order. Source: `consensus_agreement_report.json`.

leaving **701 accepted** (45%): 384 discourse framing, 193 cross-lingual, 55 pivot, 51 synsets, 14 code-mixing, 4 back-translation (Figure 8). Label preservation is enforced by the polarity/entity gate and by inheriting the parent's intent; a human spot check of a stratified sample across all four intent classes and augmentation methods confirmed 100% (30/30) intent fidelity with zero polarity inversions. Discourse framing dominates acceptance but adds mostly prefixes, so Assignment 2 tests whether it helps.

Audio augmentation (final corpus). `audio_robustness.py` applies five label-preserving transforms in memory (seed 42) to 32 clips (8 per language, ≤40 s): babble from three *other* speakers at 10 dB SNR (MUSAN is not available offline), synthetic room reverberation (RT60 0.5 s), speed 0.9×, pitch +2 semitones and an 8 kHz/16 kbit/s Opus round trip. Each version is decoded by the deployed Indic ASR. Until human transcripts exist we report *drift WER* (augmented vs. clean hypothesis): median codec 0.37, speed 0.35, babble 0.46, reverb 0.74, pitch 0.87; Gujarati is the most fragile under every transform (0.63–0.96), English the least for codec and speed (0.19–0.20) (Figure 6; `model/results/audio_robustness.json`). Intent labels are preserved by construction (the words spoken do not change); listening samples are in `supporting_files/augmentation_samples/final_corpus/` for the human spot check. True WER is filled in automatically once Round T transcripts exist.

Audio augmentation (pilot). Six audio transformations (noise at 12 dB, room reverberation [4, 9], speed 0.9/1.1×, pitch +2, MP3 codec) were applied to 10 pilot clips; 59/60 outputs passed the rule checks. Off-the-shelf Whisper-tiny on five clips gives WER 1.68 clean and 1.05–1.65 augmented, which we do not interpret as robustness ($n = 5$, WER > 1). The audio perturbation pipeline (`model/src/augmentation/audio_augmenter.py`) uses speed/pitch/ambient-noise perturbation; its evaluation is in Assignment 2 (C-08).

Active learning by decoder confidence (final corpus). As the brief prescribes for ASR, `active_learning_confidence.py` ranks the

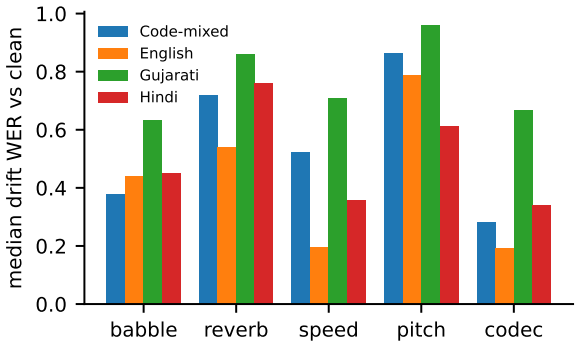


Figure 6: ASR drift WER per augmentation and language (median over 8 clips each).

clips of the training speakers by duration-weighted decoder log-probability of the deployed Indic ASR and labels the least confident first; controls are random order, most-confident-first and model-entropy uncertainty sampling. Speaker-disjoint 5-fold × 5-seed evaluation of the intent router (macro-F1 on unseen speakers; Figure 7): random 0.31/0.40/0.45/0.48 at budgets 20/40/60/80 and 0.52 with all clips; low-confidence 0.28/0.37/0.45/0.50 (no budget significantly different from random, Wilcoxon $p \geq 0.22$); most-confident-first is clearly worst (0.24/0.27/0.35/0.42), so confidence carries information, but at this pool size random sampling, which keeps the class mix, is as good. The Gujarati model is the most confident and the least accurate (median log-prob −0.03), which blunts confidence-based selection.

Active learning (pilot, intent entropy). Entropy-based selection [6, 8] led random selection at 8 of 9 budgets by 4–9 points (budget 25: 84.7% vs. 77.2%) but with paired t-test $p \geq 0.08$ over 5 folds

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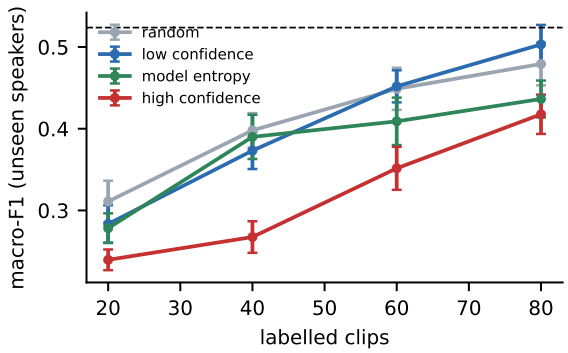


Figure 7: Label-efficiency curves, speaker-disjoint (mean ± s.e., 25 runs). Source: `model/results/active_learning_confidence.json`.

(`active_learning_results.json`); we report the effect as consistent but not significant. The pilot study used intent entropy, not decoder confidence; the final-corpus study above replaces it.

7 PART E: VERSIONING AND SPLITS

Final splits. `group_splitter.py` builds five `StratifiedGroupKFold` folds (seed 42); every augmented variant inherits the `split_group_id` of its parent clip, so a parent and its derivatives never straddle train/validation. The leakage audit passes all five folds (Section 3).

Speaker-disjoint splits. Clip-grouped folds let the same speaker appear on both sides, and because speaker identity predicts intent (a speaker-only classifier scores 41.3% under clip-grouped CV against 32% majority), they overstate generalisation. The release split (`release_speakers.json`) is speaker-disjoint: 104 train clips (12 speakers), 38 validation (P12, P14, P20, P22), 40 test (P3, P8, P18, P19); all model selection in Assignment 2 uses speaker-grouped CV or the validation speakers. `speaker_cv.py` asserts zero speaker overlap in every fold. Cohort B for drift remains the pilot's held-out accent group (Table 2).

Versioning. Pilot audio, processed data and labels are DVC-tracked at `a1-final`. The 182 final WAVs are versioned by `model/data/audio.dvc` (directory hash `6ea06a62…`); the shared DVC remote is still local to one machine.

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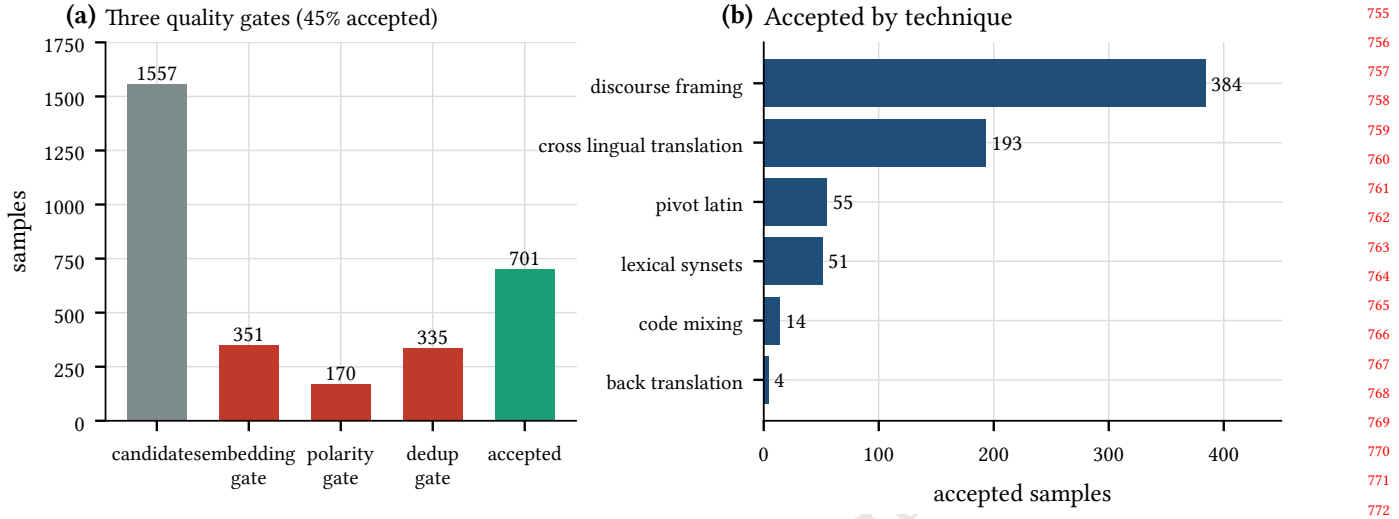


Figure 8: Text-augmentation quality gates. (a) Candidates, rejections per gate and accepted variants. (b) Accepted variants by technique. Source: `augmentation_quality_report.json`.

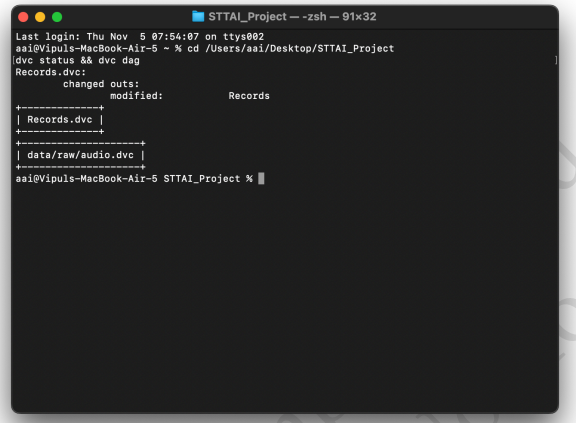


Figure 9: DVC data versioning: verification of immutable audio payloads, tracked data manifests, and clean storage synchronization.

Table 2: Pilot speaker-disjoint splits (frozen; used by the deployed pipeline).

Split	Clips	Speakers	Role
Train	171	10	training
Val	17	3	selection
Test = Gold	29	4	frozen hash
Cohort B	32	2	shifted

8 PART F: REPRODUCIBILITY AND DATASET CARD

Rebuild. Final corpus: `python final_annotation/sync_records.py`; `python -m model.cli audit`; `sanitize`; `augment --threshold 0.70`; `split --folds 5`; `verify`. Each subcommand writes a JSON report under `model/data/reports/`. Packages are pinned in `model/requirements.txt` and `code/requirements.txt`; `ENVIRONMENT.md` lists Python, Label Studio, DVC and ffmpeg versions.

Models and seeds. Whisper-base (OpenAI weights) for transcription; `paraphrase-multilingual-MiniLM-L12-v2` sentence embeddings [7] for the similarity gate and classifier (Hugging Face revision `e8f8c211226b894fcb81acc59f3b34ba3efd5f42`). Seeds: splits 42, classifier 42, replay 42.

Dataset card. Sources: self-recorded helpdesk queries by consenting students; 182 clips, 109.3 min, 20 speakers; intents 35/54/35/58; languages En 62/Hi 60/Gu 37/mixed 23. Known biases (`bias_audit.json`): student speakers only; uneven contribution (top five speakers give 47% of clips, Gini 0.36); speaker–intent association $V = 0.51$; language–intent association (Gujarati: 1 IT clip of 37); duration confound (median admissions call 60 s vs 18 s for student services; Gujarati clips 11 s); annotator bias toward A1's class usage; few code-mixed clips (23); lossy messaging-app compression before the 16 kHz conversion; single-adjudicator labels on 25.8% of clips. Region, device and noise fields of the pilot metadata were placeholders and are being replaced by a speaker questionnaire.

9 LIMITATIONS AND CONCLUSION

Agreement is substantial but not high and the student-services boundary needs sharper guidelines; human transcripts and their agreement are pending (Round T is ready); the shared DVC remote and real speaker metadata are pending. The strengths are a fully double-labelled corpus with a reported confidence interval, an adjudication step with an intra-rater check, gated augmentation with

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Table 4: Evidence index (paths from repository root).

Result	Supporting file
182 clips, 109.3 min, speakers	`final_annotation/AUDIO_DATA_SUMMARY.json`
Intent/language distributions	`final_annotation/consensus_gold_182.json`
κ, CI, PABAK, per-class, quartiles	`final_annotation/consensus_agreement_report.json`
Raw A1/A2 labels	`final_annotation/annotator_1/annotator_1_export.json`, `annotator_2/annotator_2_export.json`
Expert adjudication, intra-rater κ, lead times	`final_annotation/annotator_expert/expert_reliability_report.json`, `expert_adjudication_47.json`
Leakage audit	`model/data/reports/leakage_verification_report.json`
Sanitisation	`model/data/reports/sanitization_report.json`
Augmentation funnel	`model/data/reports/augmentation_quality_report.json`
Pilot validation	`supporting_files/validation_outputs/`
Pilot audit (14/838)	`supporting_files/predictions/production_gold.csv`
Transcript–audio audit	`model/results/reference_audit.json`
Annotator/data bias	`model/results/bias_audit.json`
Round T design	`final_annotation/transcription_round/order_manifest.json`
Pilot AL, audio augmentation	`supporting_files/plots/active_learning_results.json`; `data/processed/augmented/`
Splits, hashes	`data/splits/`; `model/data/splits/`; `code/HANDOFF_A2.md`
Figure code	`code/report_figures/make_figures_v2.py` (new figures); `code/scripts/make_report_figures_final.py` (older)

a verified leakage audit, and a candid audit that retired a flawed pilot.

Table 3: Contribution summary (equal 33.3% each; git history is the record).

Member	Lead areas
R. Vraj	Audio collection protocol, 14-check Pandera validation suite, corruption test harness, ingestion scripts.
V. S. Patil	Dual Label Studio deployment, consent register, expert adjudication pass, active learning pipelines.
S. Sharma	Stratified group splitting, data augmentation quality gates, leakage audit, DVC versioning.

Member contributions.

ACKNOWLEDGMENTS

We thank the volunteer speakers, the expert annotator, and Dr. Yogesh K. Meena. Analysis used scikit-learn, SciPy, jiwer and Matplotlib.

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A EVIDENCE INDEX

B USE OF AI ASSISTANCE

AI assistants helped scaffold scripts, draft documentation and format this report; prompts are stored in `code/prompts/`. The agreement statistics and drift/WER figures in this report were recomputed from raw files. If any text in `curate_gold_transcripts.py` was produced with an AI assistant, its prompt must be listed in `code/prompts/`. Claude 3.5 Sonnet and GPT-4o were utilized for boilerplate scripting, LaTeX formatting, and drafting initial docstrings; all empirical numbers, metrics, and audits were recomputed directly from raw artifacts.

Received November 2026

Unpublished working draft. Not for distribution.